Akka vs. PydanticAI

A comparison for teams building agentic AI

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PydanticAI gives you a clean, type-safe way to build agents in Python; Akka gives you the Akka Agentic AI Platform, which runs them and guarantees resilience and scalability. PydanticAI is a typed Python agent framework — model-agnostic, with structured outputs, dependency injection, and graph support. It is a framework, not a platform: no durable runtime, no HA/DR, no embedded governance, no operational SLA. You assemble and operate production yourself.

0	99.9999%	4	90%
PYDANTICAI FRAMEWORK SLA	AKKA PLATFORM SLA	EXTERNAL ENGINES PYDANTICAI LEANS ON	CHEAPER TO OPERATE ON AKKA

DIMENSION	PYDANTICAI	AKKA
What it is	A typed Python agent framework (library)	The Akka Agentic AI Platform, which guarantees resilience and scalability
Scope	Agent construction, structured output, tools, MCP, graph; durability, HA/DR, governance, and operations are the customer's	Orchestration, agents, memory, streaming, APIs, observability, and governance on one runtime
Durable execution	None native — wrapped around external engines (Temporal, DBOS, Prefect, Restate)	Built into the runtime; event-sourced, replayable
Availability SLA	None on the framework; SLA only on Logfire Enterprise observability	99.9999% — entire platform, backed by indemnities
RTO / RPO	Owned by the customer's chosen infrastructure	Sub-1-minute RTO; zero-byte RPO
Memory / state	Process memory; durability via the external engine's database	Durable in-memory, 4ms reads / sub-10ms writes
Governance / EU AI Act	Observability and evals via Logfire (after-the-fact); no inline enforcement or classification	Aspect-woven runtime enforcement + full pre-production governance
Real-time streaming	Not provided; sourced separately	Built in, backpressured, petabyte-scale, no external broker
Cost model	Python infrastructure footprint you provision and operate	Shared compute; up to 90% lower infrastructure for the same workload
Certifications	SOC 2 Type II, HIPAA, GDPR (Logfire)	19 standards (SOC 2 II + public SOC 3, ISO 27001/42001, HIPAA, PCI DSS, GDPR, NIS2, DORA, EU AI Act, NIST AI RMF)

PydanticAI Is a Framework; Akka Is a Platform

PydanticAI is a typed Python agent framework from the team behind Pydantic. Its design point is developer experience: model-agnostic agents, structured and validated outputs, dependency injection, MCP support, and graph-based multi-agent workflows, all checked by your type checker and IDE. For teams building agents in Python who want errors caught at write-time, that DX is genuinely clean. A framework defines how you write code; a platform runs it and guarantees it. The durable runtime, high availability, disaster recovery, streaming, the API tier, embedded governance, and the operational SLA are **not in the framework** — you source, integrate, and operate them, and own every failure across the seams.

Everything an agentic system needs runs in one Akka runtime, so a system that compiles is ready to run in production. The Akka SDK models the entire agentic system in one place — agents, tools, durable memory, streaming, APIs, and governance — and the platform generates it, tests it, and runs it as one certified service.

Capability	PydanticAI	Akka
Typed agent construction (tools, structured output, MCP, graph)	Yes — clean, type-safe	Yes
Durable runtime	Delegated to external engines	Built in
High availability / disaster recovery	Customer-owned	Built in, active-active
Durable memory	Customer-owned (external engine's DB)	Built in, 4ms / sub-10ms
Real-time streaming	Not provided	Built in, backpressured, petabyte-scale
HTTP / gRPC API layer	Not provided	Built in
Governance / policy enforcement	Observe and evaluate (Logfire)	Inline, runtime-embedded
Operational SLA	None on the framework	99.9999%, entire platform

Reliability, SLA, and Durability

PydanticAI publishes no availability SLA, because a framework does not run anything — availability is a property of the infrastructure the customer builds around it. For durability, PydanticAI leans on external durable-execution engines: it ships co-maintained integrations with **Temporal, DBOS, Prefect, and Restate**, each of which checkpoints or replays state in its own datastore. The framework documents this explicitly; durability is delegated, not native. Reliability, recovery, and the SLA for the running system are owned and operated by the customer and their chosen engines. Akka delivers durability and availability as runtime properties: state is event-sourced and replayable from the journal, recovery is automatic, and the platform carries a contractual six-nines SLA.

Metric	PydanticAI	Akka
Availability SLA	None on the framework; SLA only on Logfire Enterprise	99.9999% — entire platform
Allowed downtime / year	Whatever the assembled stack delivers	~31 seconds
RTO	Owned by the customer's chosen engine	Sub-1 minute
RPO	Owned by the customer's chosen engine	Zero byte
Durable execution	Delegated to Temporal / DBOS / Prefect / Restate	Native, in the runtime
SLA scope	None	The entire platform

Up to 90% Cheaper to Operate

AI systems built with Akka are up to **90% cheaper to operate** than Python-based systems — a function of the infrastructure required for the same agentic transaction volume, not list price. A PydanticAI deployment carries the Python infrastructure footprint, plus the separate datastore and operations of whichever durable-execution engine provides recovery, plus the memory, streaming, API, observability, and governance tiers run alongside it.

Akka runs all of it on one shared-compute runtime. The efficiency comes from actor concurrency (~10T tokens/core/year vs ~2T; ~80% less compute than Python frameworks; Manulife: up to 300% more concurrency, 30–50% faster processing after porting from Python), shared compute, and micro-checkpointing. The spend is predictable — a fixed annual fee, not a footprint that scales with load.

Maturity and Vendor

PydanticAI reached **V1 in September 2025**; the current stable release is 1.107.0 (June 2026), marked Production/Stable, with V2 in beta. The framework is young but real. Pydantic the company is venture-funded — a **$12.5M Series A led by Sequoia** (Oct 2024), ~$17.2M total — and monetizes through Logfire.

Akka is profitable and self-funding, operating since 2009 across 100,000+ production deployments, and 52 banks behind it. Dell Technologies Capital is the largest shareholder, a customer, and an AI partner — so the platform you standardize on does not depend on a venture timeline.

Continuous AI Intelligence

Akka Optimize keeps improving the AI after it ships. It runs an evaluation loop on your own production traffic, grading live interactions and training smaller specialized models on your proprietary data, so the system keeps getting better as it runs. The loop delivers three results that compound over time:

- **Model leverage** — route each request to the best model from any vendor under your policies, and reserve frontier models for the requests that need them.
- **Data sovereignty** — the specialized models are trained on your proprietary data, owned by you, and run inside your own environment.
- **Cost governance** — full visibility and control of AI spend, with savings that compound as the loop keeps running.

The evaluations run continuously inside Akka and grade traffic from agents wherever they run, in Akka or in third-party harnesses.

Governance and the EU AI Act

PydanticAI's governance story is observability. Through Pydantic Logfire — the team's commercial product — you get tracing, evaluations, and agent monitoring, with SOC 2 Type II, HIPAA, and GDPR for the observability service and an EU data region. That is real, and it matters for debugging and quality. It is observe-and-evaluate, applied to logs after execution: it does not enforce policy inline, pause or override a running process, classify a system before it ships, produce an immutable hash-chained interaction ledger, or seal an audit artifact. Those obligations remain the customer's.

The penalties are enforceable now

Violation	Maximum Fine
Prohibited AI practices (Art. 5)	EUR 35M or 7% global turnover
High-risk obligations (Art. 9-15)	EUR 15M or 3% global turnover

High-risk AI carries a 10-year logging-retention obligation (Art. 72).

How Akka governs

At the runtime: inline guardrails, policies, LLMs-as-a-judge, and sanitizers; hash-chained immutable evidence; HITL/HOTL control; classification against 240 regulations and frameworks and 1,714 controls (700 carrying a financial penalty) before a system ships; multi-persona sign-offs; a sealed Governance Posture Package; and Akka Verify proving conformance from the running system. Governance a PydanticAI team would assemble and bolt on, Akka enforces inline.

Two Lifecycles, One Certified System

Building with PydanticAI means engineers writing Python agent code; there is no built-in path for a product manager, domain expert, or risk officer to contribute, and no built-in governance lifecycle. Akka runs two independent lifecycles on one platform via **Akka Specify**:

BUILD LIFECYCLE

FUNCTIONAL CONTRACT

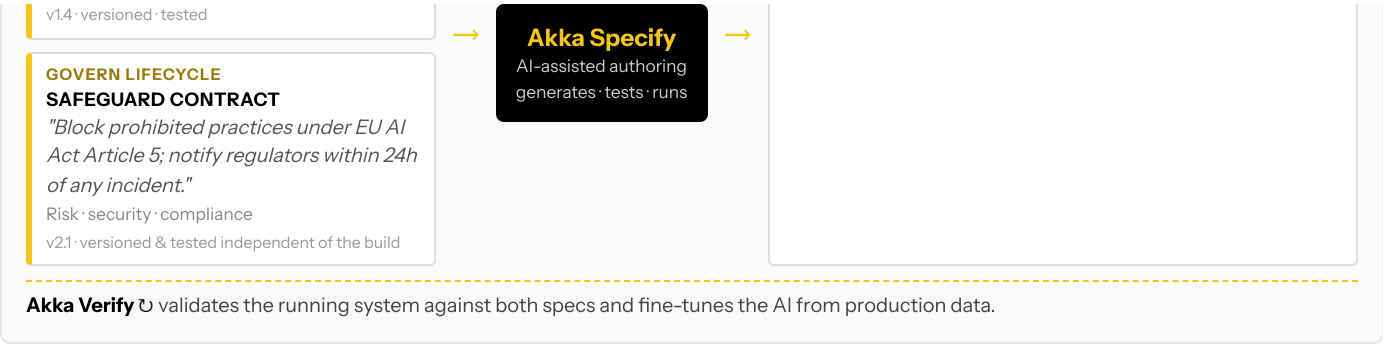
"Rank incoming ER patients by acuity and route the top three to a clinician."

Product · developers · ML engineers · domain experts

ONE CERTIFIED AI SERVICE

Built, governed, and running

- Agents, tools, orchestration, memory, APIs, streaming, UI
- Guardrails, sanitizers, HITL/HOTL, evaluations, halts
- Interaction, evidence, and causal logging



The build lifecycle and the governance lifecycle are versioned and tested independently, by different audiences — an audience and a workflow PydanticAI has no equivalent for.

Real-Time Streaming at Petabyte Scale

PydanticAI has no streaming engine; real-time pipelines are provisioned separately. Akka's streaming is built into the runtime — continuous, backpressured, **petabyte-scale, in-memory**, with no external broker — powering both agent feedback loops and high-throughput data processing (the engine behind Tubi's real-time hyper-personalization at 5 billion tokens per second).

For the Buyer: Risk, Compliance, and Accountability

Buyer concern	PydanticAI	Akka
Certifications & audits	SOC 2 Type II, HIPAA, GDPR (Logfire observability)	19 standards — SOC 2 II + public SOC 3, ISO 27001/42001, HIPAA, PCI DSS, GDPR, NIS2, DORA, EU AI Act, NIST AI RMF — plus annual pen tests, SBOMs, 40+ policies (trust.akka.io)
Scope of accountability	The framework; you integrate and operate the runtime, durability, and everything else	One platform, one SLA, 24/7 SRE — Akka owns the running system
Risk transfer	Standard open-source / cloud terms	Availability and data-integrity guarantees backed by contractual indemnities
Track record & funding model	Venture-funded: $12.5M Series A led by Sequoia (Oct 2024), ~$17.2M total; monetizes via Logfire	Profitable and self-funding; since 2009 and 100,000+ deployments (52 banks); Dell Technologies Capital is largest shareholder, a customer, and an AI partner
Budget predictability	Python infrastructure footprint you size and operate	Fixed annual fee finance can forecast

PydanticAI gives you a clean, type-safe way to author agents in Python. Akka gives you the platform that runs them: durable runtime, high availability, embedded governance, and an SLA Akka owns. The decision is scope and accountability — a framework you build a production system around, versus the production system.

Customers Running Agentic and Real-Time Systems on Akka

Manulife
2,000
developers across 100 projects on one governed platform

Tubi
5B tok/s
real-time hyper-personalization engine

Swiggy
71ms
order-assignment AI, ~50% latency reduction

John Deere
1,000+
tractor sensors turned into real-time insight

Verizon
750%
order-processing capacity gain; 6s → 2.4s response

Common Questions

We like PydanticAI's typed developer experience. Why add Akka?

PydanticAI's type-safe DX is a real strength for authoring agents in Python. It is a framework, so the durable runtime, high availability, disaster recovery, streaming, the API tier, embedded governance, and the operational SLA are yours to build and run. Akka delivers those as one platform with a 99.9999% SLA, and teams can keep their typed agent design discipline while moving production onto a runtime that guarantees it.

Can we add governance on top of PydanticAI with Logfire?

Logfire gives you tracing and evaluations — observe-and-evaluate on logs after execution. The EU AI Act expects enforcement inline to the runtime: immutable records witnessed as they happen, human override on running processes, pre-deployment classification, and a sealed audit artifact. Bolt-on observability cannot gate a deployment or classify a system before it ships. Akka embeds inline enforcement and covers pre-deployment governance.

PydanticAI supports durable execution. Doesn't that cover reliability?

PydanticAI integrates with external durable-execution engines — Temporal, DBOS, Prefect, and Restate — and its own docs frame durability as delegated to those systems. You select, provision, and operate a separate engine and its datastore, and own the SLA, HA/DR, and recovery across the seams. Akka's durability is native to the runtime: event-sourced, replayable, sub-1-minute RTO, zero-byte RPO, under a six-nines SLA.

Is PydanticAI cheaper because it is open source?

The framework is open source, but production means the Python infrastructure footprint plus a separate durable-execution engine and the memory, streaming, API, observability, and governance tiers you provision and operate. Akka's shared-compute model is up to 90% cheaper to operate for the same agentic transaction volume, on a fixed annual fee.

Sources

PydanticAI — what it is: ai.pydantic.dev, pydantic.dev/pydantic-ai — model-agnostic, type-safe Python agent framework; structured outputs, dependency injection, MCP, graph workflows

PydanticAI durable execution: ai.pydantic.dev/durable_execution/overview — co-maintained integrations with Temporal, DBOS, Prefect, Restate; durability delegated to external engines

PydanticAI maturity: github.com/pydantic/pydantic-ai/releases, pypi.org/project/pydantic-ai — V1 since Sept 2025; stable 1.107.0 (June 10, 2026); V2 in beta; "Production/Stable"

Pydantic Logfire: pydantic.dev/logfire, logfire.pydantic.dev/docs/compliance — tracing and evals; SOC 2 Type II, HIPAA, GDPR, EU data region; SLA on Enterprise plan

Pydantic funding: techcrunch.com (Oct 1, 2024), crunchbase.com — $12.5M Series A led by Sequoia (Oct 2024); ~$17.2M total over two rounds

Akka trust center: trust.akka.io — 19 compliance standards; SOC 2 II + public SOC 3; annual pen tests, SBOMs, 40+ policies

Akka performance: akka.io/blog/go-slow-to-go-fast — Manulife up to 300% more concurrency, 30–50% faster; ~10T vs ~2T tokens/core; ~80% less compute than Python

Akka platform: 99.9999% availability, active-active HA/DR, sub-1 min RTO, zero byte RPO (contractual indemnities); 240 regulations and frameworks / 1,714 controls / 700 with financial penalty; 100,000+ deployments / since 2009 / 52 banks; profitable; Dell Technologies Capital